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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 29th July Evening Shift

JEE Main Question Paper with Answer Key

29 Jul 2022	90	22	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	21 questions
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Out of syllabus	22 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

Consider the reaction

$4HNO_3(l) + 3KCl(s) \rightarrow Cl_2(g) + NOCl(g) + 2H_2O(g) + 3KNO_3(s)$

The amount of HNO_3 required to produce $110.0\ g$ of KNO_3 is

(Given: Atomic masses of H, O, N and K are $1, 16, 14$ and 39, respectively.)

(A) 32.2 g (B) 69.4 g (C) 91.5 g (D) 162.5 g

Open solution

Q2. MCQ (Single Correct Answer)

Given below are the quantum numbers for 4 electrons.

A. $n = 3, l = 2, m_l = 1, m_s = +1/2$

B. $n = 4, l = 1, m_l = 0, m_s = +1/2$

C. $n = 4, l = 2, m_l = -2, m_s = -1/2$

D. $n = 3, l = 1, m_l = -1, m_s = +1/2$

The correct order of increasing energy is :

(A) D < B < A < C (B) D < A < B < C (C) B < D < A < C (D) B < D < C < A

Open solution

Q3. MCQ (Single Correct Answer)

$C(s) + O_2(g) \rightarrow CO_2(g) + 400\ kJ$

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) + 100\ kJ$

When coal of purity 60% is allowed to burn in presence of insufficient oxygen, 60% of carbon is converted into 'CO' and the remaining is converted into 'CO_2'. The heat generated when $0.6\ kg$ of coal is burnt is _____.

(A) 1600 kJ (B) 3200 kJ (C) 4400 kJ (D) 6600 kJ

Open solution

Q4. MCQ (Single Correct Answer)

$200\ mL$ of $0.01\ M\ HCl$ is mixed with $400\ mL$ of $0.01\ M\ H_2SO_4$. The pH of the mixture is _____.

Given: $\log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.70, \log 7 = 0.84, \log 11 = 1.04$

(A) 1.14 (B) 1.78 (C) 2.34 (D) 3.02

Open solution

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Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements.

Statement I : Stannane is an example of a molecular hydride.

Statement II : Stannane is a planar molecule.

In the light of the above statement, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q6. MCQ (Single Correct Answer)

Open solution

Which of the following 3d-metal ion will give the lowest enthalpy of hydration $\left(\Delta_{\text{hyd}} \mathrm{H}\right)$ when dissolved in water ?

(A) Cr^{2+} (B) Mn^{2+} (C) Fe^{2+} (D) Co^{2+}

Q7. MCQ (Single Correct Answer)

Open solution

Octahedral complexes of copper(II) undergo structural distortion (Jahn-Teller). Which one of the given copper (II) complexes will show the maximum structural distortion? (en - ethylenediamine; $\mathrm{H_2N-CH_2-CH_2-NH_2}$)

(A) $[\mathrm{Cu(H_2O)_6}]\mathrm{SO_4}$

(B) $[\mathrm{Cu(en)(H_2O)_4}]\mathrm{SO_4}$

(C) cis-$[\mathrm{Cu\,(en)_2Cl_2}]$

(D) trans-$[\mathrm{Cu\,(en)_2Cl_2}]$

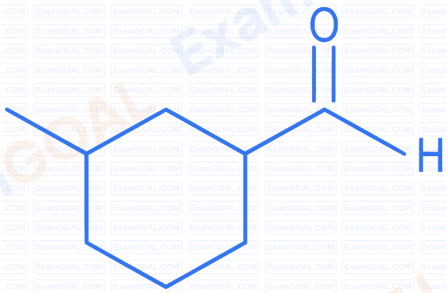
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Q8. MCQ (Single Correct Answer)

Open solution

Correct structure of γ-methylcyclohexane carbaldehyde is

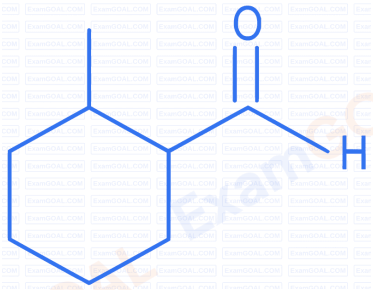
(A)



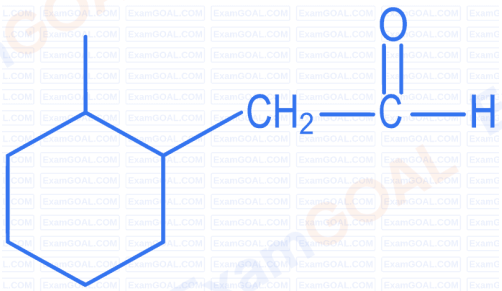
(B)



(C)



(D)



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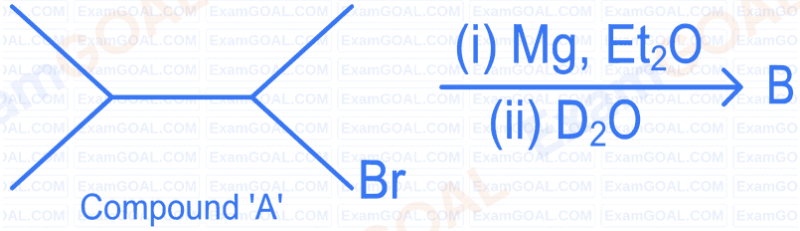
Q9. MCQ (Single Correct Answer)

Open solution

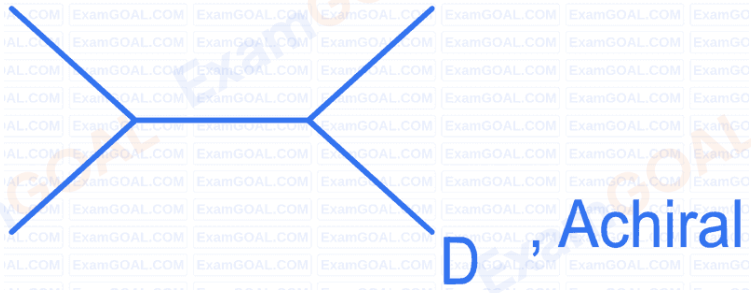
Compound 'A' undergoes following sequence of reactions to give compound 'B'.

The correct structure and chirality of compound 'B' is

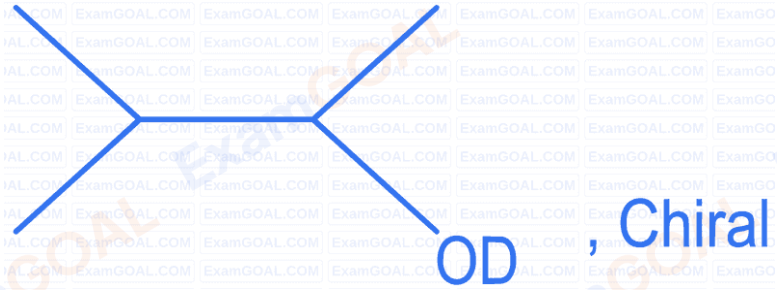
[where Et is $-C_2H_5$]



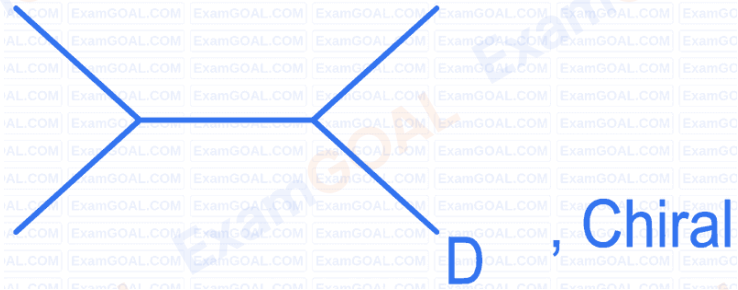
(A)



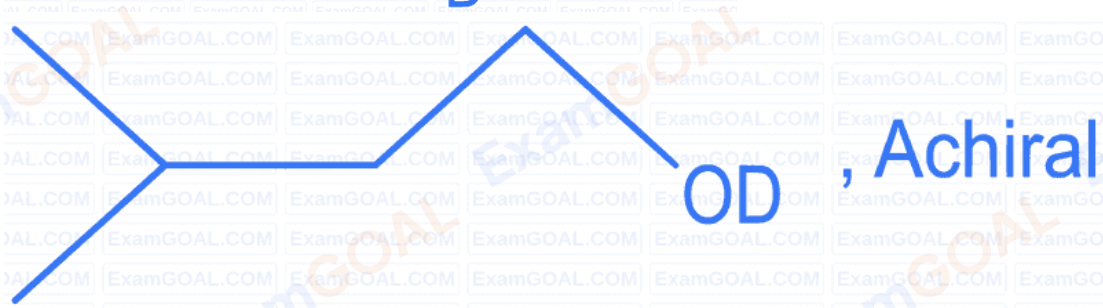
(B)



(C)



(D)

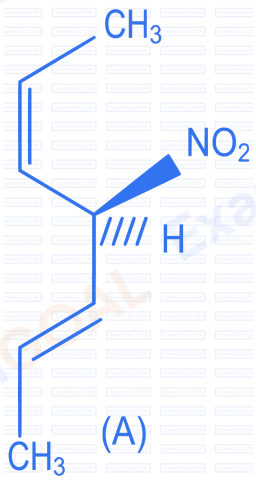


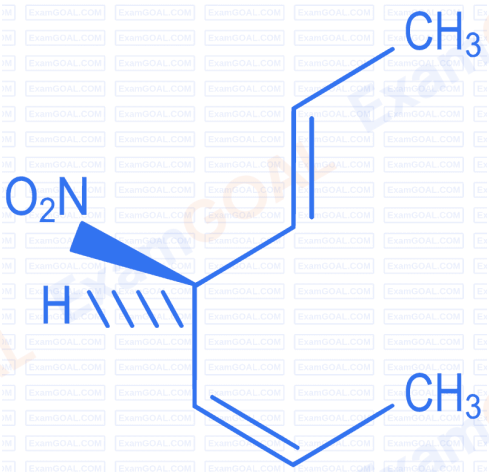
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Q10. MCQ (Single Correct Answer)

Open solution

Given below are two statements.

Statement I : The compound  is optically active.

Statement II :  is mirror image of above compound A.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q11. MCQ (Single Correct Answer)

Open solution

When ethanol is heated with conc. H_2SO_4, a gas is produced. The compound formed, when this gas is treated with cold dilute aqueous solution of Baeyer's reagent, is

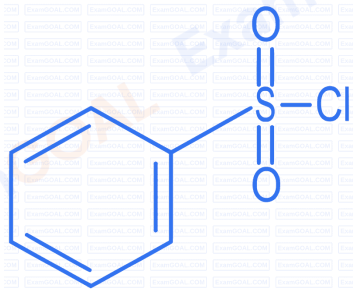
(A) formaldehyde (B) formic acid (C) glycol (D) ethanoic acid

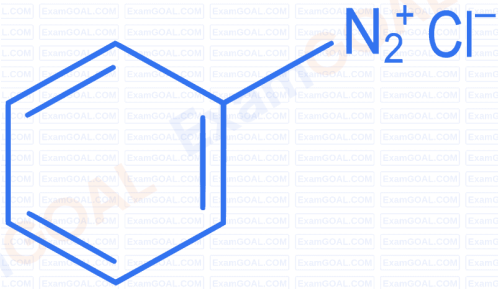
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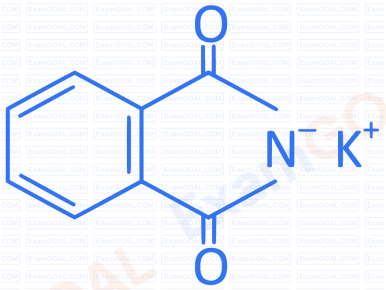
Q12. MCQ (Single Correct Answer)

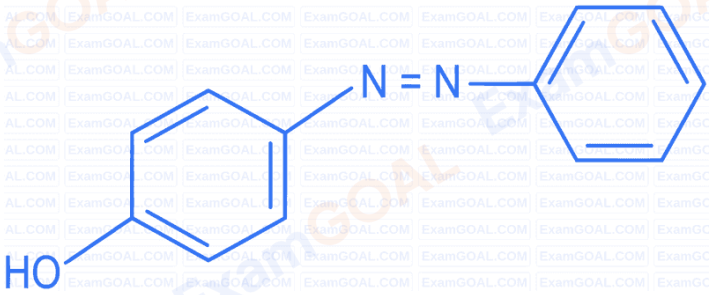
Open solution

The Hinsberg reagent is

(A) 

(B) 

(C) 

(D) 

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Q13. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Amylose is insoluble in water.

Reason R: Amylose is a long linear molecule with more than 200 glucose units.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

(C) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

(D) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct.

Open solution

Q14. MCQ (Single Correct Answer)

A compound 'X' is a weak acid and it exhibits colour change at pH close to the equivalence point during neutralization of NaOH with CH_3COOH. Compound 'X' exists in ionized form in basic medium. The compound 'X' is

(A) methyl orange

(B) methyl red

(C) phenolphthalein

(D) erichrome Black T

Open solution

Q15. Numerical

Consider, $PF_5, BrF_5, PCl_3, SF_6, [ICl_4]^-, ClF_3$ and IF_5.

Amongst the above molecule(s)/ion(s), the number of molecule(s)/ion(s) having $sp^3\,d^2$ hybridisation is _____.

Open solution

Q16. Numerical

$1.80\ g$ of solute A was dissolved in $62.5\ cm^3$ of ethanol and freezing point of the solution was found to be $155.1\ K$. The molar mass of solute A is _____ g mol^{-1}.

[Given : Freezing point of ethanol is 156.0 K.

Density of ethanol is 0.80 g cm^{-3}.

Freezing point depression constant of ethanol is 2.00 K kg mol^{-1}]

Open solution

Q17. Numerical

For a cell, $Cu(s)\,|Cu^{2+}(0.001\ M)||Ag^+(0.01\ M)|Ag(s)$

the cell potential is found to be $0.43\ V$ at $298\ K$. The magnitude of standard electrode potential for Cu^{2+}/Cu is _____ $\times 10^{-2}$ V.

[Given : $E^{\Theta}_{Ag^+/Ag}$ = 0.80 V and $\frac{2.303RT}{F}$ = 0.06 V]

Open solution

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Q18. Numerical

Assuming $1\ \mu g$ of trace radioactive element X with a half life of 30 years is absorbed by a growing tree. The amount of X remaining in the tree after 100 years is _____ $\times 10^{-1} \mu g$.

[Given : ln 10 = 2.303; log 2 = 0.30]

Open solution

Q19. Numerical

Sum of oxidation state (magnitude) and coordination number of cobalt in $\mathrm{Na}\left[\mathrm{Co(bpy)Cl_4}\right]$ is _____.

(Given : bpy = )

Open solution

Q20. Numerical

A 1.84 mg sample of polyhydric alcoholic compound 'X' of molar mass 92.0 g/mol gave 1.344 mL of H_2 gas at STP. The number of alcoholic hydrogens present in compound 'X' is _____.

Open solution

Q21. Numerical

The number of stereoisomers formed in a reaction of $(\pm)\mathrm{Ph(C=O)C(OH)(CN)Ph}$ with HCN is _____.

[where Ph is $-\mathrm{C_6H_5}$]

Open solution

Mathematics 21 questions

Q22. MCQ (Single Correct Answer)

If $z \neq 0$ be a complex number such that $\left|z - \frac{1}{z}\right| = 2$, then the maximum value of $|z|$ is :

(A) $\sqrt{2}$ (B) 1 (C) $\sqrt{2}-1$ (D) $\sqrt{2}+1$

Open solution

Q23. MCQ (Single Correct Answer)

Which of the following matrices can NOT be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation ?

(A) $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$

(D) $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Open solution

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Q24. MCQ (Single Correct Answer)

If the system of equations

$x + y + z = 6$

$2x + 5y + \alpha z = \beta$

$x + 2y + 3z = 14$

has infinitely many solutions, then $\alpha + \beta$ is equal to

(A) 8 (B) 36 (C) 44 (D) 48

Open solution

Q25. MCQ (Single Correct Answer)

Let the function $f(x) = \begin{cases} \frac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{ if } x \neq 0 \\ 10 & ; \text{ if } x = 0 \end{cases}$ be continuous at $x = 0$.

Then α is equal to

(A) 10 (B) -10 (C) 5 (D) -5

Open solution

Q26. MCQ (Single Correct Answer)

If $[t]$ denotes the greatest integer $\leq t$, then the value of $\int_0^1 \left[2x - \left|3x^2 - 5x + 2\right| + 1\right] dx$ is :

(A) $\frac{\sqrt{37}+\sqrt{13}-4}{6}$ (B) $\frac{\sqrt{37}-\sqrt{13}-4}{6}$ (C) $\frac{-\sqrt{37}-\sqrt{13}+4}{6}$ (D) $\frac{-\sqrt{37}+\sqrt{13}+4}{6}$

Open solution

Q27. MCQ (Single Correct Answer)

For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$, if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

(A) $3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ (B) $3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

(C) $3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ (D) $3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

Open solution

Q28. MCQ (Single Correct Answer)

If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passes through the points $(2, 1)$ and $(k + 1, 2), k > 0$, then

(A) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2 + 1\right)$ (B) $\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2 + 1\right)$

(C) $2\tan^{-1}\left(\frac{1}{k+1}\right) = \log_e\left(k^2 + 2k + 2\right)$ (D) $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(\frac{k^2+1}{k^2}\right)$

Open solution

Q29. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y = \frac{(x+3)}{x+1}, x > -1$, which passes through the point $(0, 1)$. Then $y(1)$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Open solution

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Q30. MCQ (Single Correct Answer)

Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3\left(m_1^2 + m_2^2\right) = 220$. If one vertex of the square is $(10(\cos\alpha - \sin\alpha), 10(\sin\alpha + \cos\alpha))$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha - \sin\alpha)x + (\sin\alpha + \cos\alpha)y = 10$, then $72\left(\sin^4\alpha + \cos^4\alpha\right) + a^2 - 3a + 13$ is equal to :

(A) 119 (B) 128 (C) 145 (D) 155

Open solution

Q31. MCQ (Single Correct Answer)

Let $\mathrm{A}(\alpha, -2), \mathrm{B}(\alpha, 6)$ and $\mathrm{C}\left(\frac{\alpha}{4}, -2\right)$ be vertices of a $\triangle\mathrm{ABC}$. If $\left(5, \frac{\alpha}{4}\right)$ is the circumcentre of $\triangle\mathrm{ABC}$, then which of the following is NOT correct about $\triangle\mathrm{ABC}$?

(A) area is 24 (B) perimeter is 25

(C) circumradius is 5 (D) inradius is 2

Open solution

Q32. MCQ (Single Correct Answer)

Bag I contains 3 red, 4 black and 3 white balls and Bag II contains 2 red, 5 black and 2 white balls. One ball is transferred from Bag I to Bag II and then a ball is drawn from Bag II. The ball so drawn is found to be black in colour. Then the probability, that the transferred ball is red, is :

(A) $\frac{4}{9}$ (B) $\frac{5}{18}$ (C) $\frac{1}{6}$ (D) $\frac{3}{10}$

Open solution

Q33. MCQ (Single Correct Answer)

Let $\mathrm{S} = \{z = x + iy : |z - 1 + i| \geq |z|, |z| < 2, |z + i| = |z - 1|\}$. Then the set of all values of x, for which $w = 2x + iy \in \mathrm{S}$ for some $y \in \mathbb{R}$, is :

(A) $\left(-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$ (B) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$

(C) $\left(-\sqrt{2}, \frac{1}{2}\right]$ (D) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

Open solution

Q34. MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar concurrent vectors such that the angle between any two of them is the same. If the product of their magnitudes is 14 and $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = 168$, then $|\vec{a}| + |\vec{b}| + |\vec{c}|$ is equal to :

(A) 10 (B) 14 (C) 16 (D) 18

Open solution

Q35. MCQ (Single Correct Answer)

The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2 - 3x + 2}{x^2 + 2x + 7}\right)$ is :

(A) $[1, \infty)$ (B) $[-1, 2]$ (C) $[-1, \infty)$ (D) $(-\infty, 2]$

Open solution

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Q36. Numerical

Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in \mathrm{N}$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____.

Open solution

Q37. Numerical

Let $X = \begin{bmatrix}1\\1\\1\end{bmatrix}$ and $A = \begin{bmatrix}-1 & 2 & 3\\0 & 1 & 6\\0 & 0 & -1\end{bmatrix}$. For $\mathrm{k} \in N$, if $X'A^kX = 33$, then k is equal to _____.

Open solution

Q38. Numerical

The number of natural numbers lying between 1012 and 23421 that can be formed using the digits $2, 3, 4, 5, 6$ (repetition of digits is not allowed) and divisible by 55 is _____.

Open solution

Q39. Numerical

If $[t]$ denotes the greatest integer $\leq t$, then the number of points, at which the function $f(x) = 4|2x + 3| + 9\left[x + \frac{1}{2}\right] - 12[x + 20]$ is not differentiable in the open interval $(-20, 20)$, is _____.

Open solution

Q40. Numerical

Let AB be a chord of length 12 of the circle $(x - 2)^2 + (y + 1)^2 = \frac{169}{4}$. If tangents drawn to the circle at points A and B intersect at the point P, then five times the distance of point P from chord AB is equal to _____.

Open solution

Q41. Numerical

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2|\vec{b}|^2, \vec{a} \cdot \vec{b} = 3$ and $|\vec{a} \times \vec{b}|^2 = 75$. Then $|\vec{a}|^2$ is equal to _____.

Open solution

Q42. Numerical

Let $S = \{(x, y) \in \mathbb{N} \times \mathbb{N} : 9(x - 3)^2 + 16(y - 4)^2 \leq 144\}$ and $T = \{(x, y) \in \mathbb{R} \times \mathbb{R} : (x - 7)^2 + (y - 4)^2 \leq 36\}$. Then $n(S \cap T)$ is equal to _____.

Open solution

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Physics 26 questions

Q43. MCQ (Single Correct Answer)

Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F. Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be:

Open solution

(A) 3F/2 (B) 3F/4 (C) F (D) 2F

Q44. MCQ (Single Correct Answer)

Match List I with List II.

	List I		List II
A.	Torque	I.	Nms^{-1}
B.	Stress	II.	$\mathrm{J\ kg}^{-1}$
C.	Latent Heat	III.	Nm
D.	Power	IV.	Nm^{-2}

Open solution

Choose the correct answer from the options given below :

(A) A-III, B-II, C-I, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-IV, B-I, C-III, D-II

(D) A-II, B-III, C-I, D-IV

Q45. MCQ (Single Correct Answer)

Two identical thin metal plates have charges q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C. The potential difference between them is :

Open solution

(A) $\frac{(q_1+q_2)}{C}$

(B) $\frac{(q_1-q_2)}{C}$

(C) $\frac{(q_1-q_2)}{2C}$

(D) $\frac{2(q_1-q_2)}{C}$

Q46. MCQ (Single Correct Answer)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Open solution

Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.

Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

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Q47. MCQ (Single Correct Answer)

A $1\ \mathrm{m}$ long wire is broken into two unequal parts X and Y. The X part of the wire is streched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of X and Y.

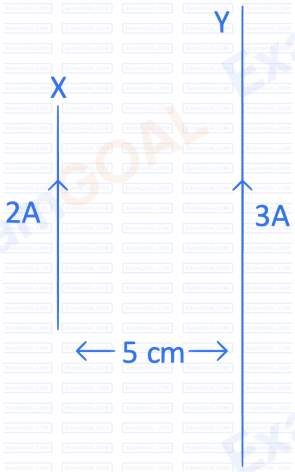
Open solution

(A) 1 : 4 (B) 1 : 2 (C) 4 : 1 (D) 2 : 1

Q48. MCQ (Single Correct Answer)

A wire X of length $50\ \mathrm{cm}$ carrying a current of $2\ \mathrm{A}$ is placed parallel to a long wire Y of length $5\ \mathrm{m}$. The wire Y carries a current of $3\ \mathrm{A}$. The distance between two wires is $5\ \mathrm{cm}$ and currents flow in the same direction. The force acting on the wire Y is

Open solution



(A) $1.2 \times 10^{-5}\ \mathrm{N}$ directed towards wire X.

(B) $1.2 \times 10^{-4}\ \mathrm{N}$ directed away from wire X.

(C) $1.2 \times 10^{-4}\ \mathrm{N}$ directed towards wire X.

(D) $2.4 \times 10^{-5}\ \mathrm{N}$ directed towards wire X.

Q49. MCQ (Single Correct Answer)

A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

Open solution

(A) g/2n (B) g/n (C) 2gn (D) $g/2n^2$

Q50. MCQ (Single Correct Answer)

A circuit element X when connected to an a.c. supply of peak voltage $100\ \mathrm{V}$ gives a peak current of $5\ \mathrm{A}$ which is in phase with the voltage. A second element Y when connected to the same a.c. supply also gives the same value of peak current which lags behind the voltage by $\frac{\pi}{2}$. If X and Y are connected in series to the same supply, what will be the rms value of the current in ampere?

Open solution

(A) $\frac{10}{\sqrt{2}}$ (B) $\frac{5}{\sqrt{2}}$ (C) $5\sqrt{2}$ (D) $\frac{5}{2}$

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Q51. MCQ (Single Correct Answer)

An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is

(A) $\frac{I_0}{4}$ (B) $\frac{I_0}{2}$ (C) $\frac{3I_0}{4}$ (D) $\frac{3I_0}{2}$

Open solution

Q52. MCQ (Single Correct Answer)

An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particles will be:

(A) $\sqrt{2}:1$ (B) $2\sqrt{2}:1$ (C) $4\sqrt{2}:1$ (D) $8:1$

Open solution

Q53. MCQ (Single Correct Answer)

An object of mass $1\ \text{kg}$ is taken to a height from the surface of earth which is equal to three times the radius of earth. The gain in potential energy of the object will be [If, $g = 10\ \text{ms}^{-2}$ and radius of earth $= 6400\ \text{km}$]

(A) 48 MJ (B) 24 MJ (C) 36 MJ (D) 12 MJ

Open solution

Q54. MCQ (Single Correct Answer)

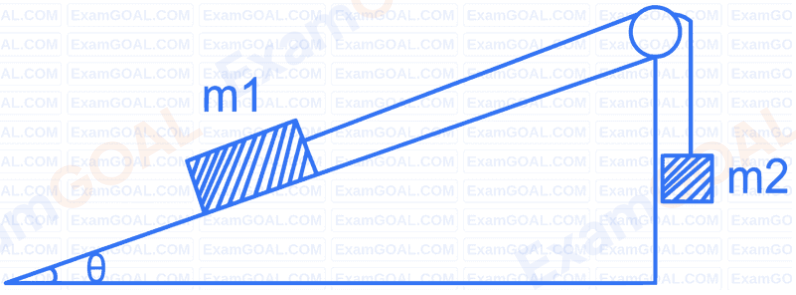
A ball is released from a height h. If t_1 and t_2 be the time required to complete first half and second half of the distance respectively. Then, choose the correct relation between t_1 and t_2.

(A) $t_1 = (\sqrt{2})t_2$ (B) $t_1 = (\sqrt{2}-1)t_2$ (C) $t_2 = (\sqrt{2}+1)t_1$ (D) $t_2 = (\sqrt{2}-1)t_1$

Open solution

Q55. MCQ (Single Correct Answer)

Two bodies of masses $m_1 = 5\ \text{kg}$ and $m_2 = 3\ \text{kg}$ are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be : [Take $g = 10\ \text{ms}^{-2}$]



(A) 30 N (B) 40 N (C) 50 N (D) 60 N

Open solution

Q56. MCQ (Single Correct Answer)

If momentum of a body is increased by 20%, then its kinetic energy increases by

(A) 36% (B) 40% (C) 44% (D) 48%

Open solution

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Q57. MCQ (Single Correct Answer)

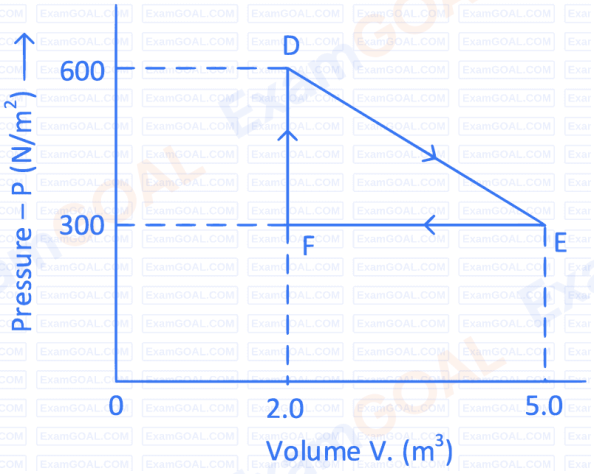
The torque of a force $5\hat{i}+3\hat{j}-7\hat{k}$ about the origin is τ. If the force acts on a particle whose position vector is $2\hat{i}+2\hat{j}+\hat{k}$, then the value of τ will be

(A) $11\hat{i}+19\hat{j}-4\hat{k}$ (B) $-11\hat{i}+9\hat{j}-16\hat{k}$ (C) $-17\hat{i}+19\hat{j}-4\hat{k}$ (D) $17\hat{i}+9\hat{j}+16\hat{k}$

Open solution

Q58. MCQ (Single Correct Answer)

A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will be



(A) -450 J (B) 450 J (C) 900 J (D) 1350 J

Open solution

Q59. MCQ (Single Correct Answer)

The root mean square speed of smoke particles of mass $5\times10^{-17}\,\mathrm{kg}$ in their Brownian motion in air at NTP is approximately. [Given $\mathrm{k}=1.38\times10^{-23}\mathrm{JK}^{-1}$]

(A) $60\,\mathrm{mm\,s^{-1}}$ (B) $12\,\mathrm{mm\,s^{-1}}$ (C) $15\,\mathrm{mm\,s^{-1}}$ (D) $36\,\mathrm{mm\,s^{-1}}$

Open solution

Q60. MCQ (Single Correct Answer)

Light enters from air into a given medium at an angle of 45° with interface of the air-medium surface. After refraction, the light ray is deviated through an angle of 15° from its original direction. The refractive index of the medium is:

(A) 1.732 (B) 1.333 (C) 1.414 (D) 2.732

Open solution

Q61. Numerical

A tube of length $50\,\mathrm{cm}$ is filled completely with an incompressible liquid of mass $250\,\mathrm{g}$ and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F}\,\mathrm{rad\,s^{-1}}$. If F be the force exerted by the liquid at the other end then the value of x will be _____.

Open solution

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Q62. Numerical

Nearly 10% of the power of a $110\ \mathrm{W}$ light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of $1\ \mathrm{m}$ from the bulb to a distance of $5\ \mathrm{m}$ is $a \times 10^{-2}\ \mathrm{W/m^2}$. The value of 'a' will be _____.

Open solution

Q63. Numerical

A metal wire of length $0.5\ \mathrm{m}$ and cross-sectional area $10^{-4}\ \mathrm{m^2}$ has breaking stress $5 \times 10^{8}\ \mathrm{Nm^{-2}}$. A block of $10\ \mathrm{kg}$ is attached at one end of the string and is rotating in a horizontal circle. The maximum linear velocity of block will be _____ $\mathrm{ms^{-1}}$.

Open solution

Q64. Numerical

The velocity of a small ball of mass $0.3\ \mathrm{g}$ and density $8\ \mathrm{g/cc}$ when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $1.3\ \mathrm{g/cc}$, then the value of viscous force acting on the ball will be $x \times 10^{-4}\ \mathrm{N}$. The value of x is _____. [use $\mathrm{g} = 10\ \mathrm{m/s^2}$]

Open solution

Q65. Numerical

The speed of a transverse wave passing through a string of length $50\ \mathrm{cm}$ and mass $10\ \mathrm{g}$ is $60\ \mathrm{ms^{-1}}$. The area of cross-section of the wire is $2.0\ \mathrm{mm^2}$ and its Young's modulus is $1.2 \times 10^{11}\mathrm{Nm^{-2}}$. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}\ \mathrm{m}$. The value of x is _____.

Open solution

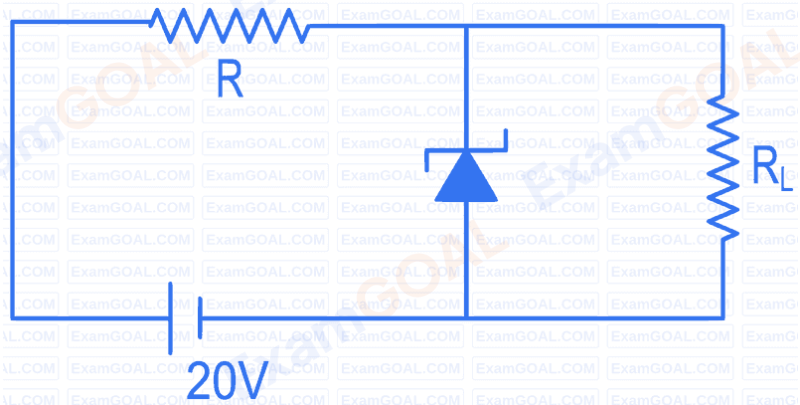
Q66. Numerical

The metallic bob of simple pendulum has the relative density 5. The time period of this pendulum is $10\ \mathrm{s}$. If the metallic bob is immersed in water, then the new time period becomes $5\sqrt{x}$ s. The value of x will be _____.

Open solution

Q67. Numerical

A $8\ \mathrm{V}$ Zener diode along with a series resistance R is connected across a $20\ \mathrm{V}$ supply (as shown in the figure). If the maximum Zener current is $25\ \mathrm{mA}$, then the minimum value of R will be _____ Ω.



Open solution

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Q68. Numerical

A capacitor of capacitance 500 μF is charged completely using a dc supply of 100 V. It is now connected to an inductor of inductance 50 mH to form an LC circuit. The maximum current in LC circuit will be _____ A.

Open solution

Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q69. MCQ (Single Correct Answer) · **Chemistry**

Given below are the critical temperatures of some of the gases:

Gas	Critical temperature (K)
He	5.2
CH_4	190.0
CO_2	304.2
NH_3	405.5

The gas showing least adsorption on a definite amount of charcoal is

(A) He (B) CH_4 (C) CO_2 (D) NH_3

Open solution

Q70. MCQ (Single Correct Answer) · **Chemistry**

In liquation process used for tin (Sn), the metal :

(A) is reacted with acid.

(B) is dissolved in water.

(C) is brought to molten form which is made to flow on a slope.

(D) is fused with NaOH

Open solution

Q71. MCQ (Single Correct Answer) · **Chemistry**

Portland cement contains 'X' to enhance the setting time. What is 'X'?

(A) $CaSO_4 \cdot \frac{1}{2}H_2O$ (B) $CaSO_4 \cdot 2H_2O$

(C) $CaSO_4$ (D) $CaCO_3$

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

When borax is heated with CoO on a platinum loop, blue coloured bead formed is largely due to :

(A) B_2O_3 (B) $Co(BO_2)_2$

(C) CoB_4O_7 (D) $Co\,[B_4O_5(OH)_4]$

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

Dinitrogen is a robust compound, but reacts at high altitudes to form oxides. The oxide of nitrogen that can damage plant leaves and retard photosynthesis is :

(A) NO (B) NO_3^- (C) NO_2 (D) NO_2^-

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Which of the following is NOT a natural polymer?

(A) Protein (B) Starch (C) Rubber (D) Rayon

Open solution

Q75. Numerical · **Chemistry**

'x' g of molecular oxygen (O_2) is mixed with $200\ g$ of neon (Ne). The total pressure of the non-reactive mixture of O_2 and Ne in the cylinder is 25 bar. The partial pressure of Ne is 20 bar at the same temperature and volume. The value of 'x' is _____.

[Given: Molar mass of $O_2 = 32\ g\ mol^{-1}$.

Molar mass of $Ne = 20\ g\ mol^{-1}$]

Open solution

Q76. Numerical · **Chemistry**

Consider the following sulphur based oxoacids.

$H_2SO_3, H_2SO_4, H_2S_2O_8$ and $H_2S_2O_7$.

Amongst these oxoacids, the number of those with peroxo $(O-O)$ bond is _____.

Open solution

Q77. Numerical · **Chemistry**

The number of chlorine atoms in bithionol is _____.

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and $a_{n+2} = 3a_{n+1} - 2a_n + 1, \forall n \geq 0$.

Then $a_{25}a_{23} - 2a_{25}a_{22} - 2a_{23}a_{24} + 4a_{22}a_{24}$ is equal to

(A) 483 (B) 528 (C) 575 (D) 624

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

$\sum_{r=1}^{20} (r^2+1)(r!)$ is equal to

(A) $22! - 21!$ (B) $22! - 2(21!)$ (C) $21! - 2(20!)$ (D) $21! - 20!$

Open solution

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Q80. MCQ (Single Correct Answer) · **Mathematics**

The number of elements in the set $S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}\right\}$ is :

(A) 1 (B) 3 (C) 0 (D) infinite

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let Q be the foot of perpendicular drawn from the point $P(1, 2, 3)$ to the plane $x + 2y + z = 14$. If R is a point on the plane such that $\angle PRQ = 60^\circ$, then the area of $\triangle PQR$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) 3

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

If $(2, 3, 9), (5, 2, 1), (1, \lambda, 8)$ and $(\lambda, 2, 3)$ are coplanar, then the product of all possible values of λ is:

(A) $\frac{21}{2}$ (B) $\frac{59}{8}$ (C) $\frac{57}{8}$ (D) $\frac{95}{8}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The statement $(p \Rightarrow q) \vee (p \Rightarrow r)$ is NOT equivalent to

(A) $(p \wedge (\sim r)) \Rightarrow q$ (B) $(\sim q) \Rightarrow ((\sim r) \vee p)$

(C) $p \Rightarrow (q \vee r)$ (D) $(p \wedge (\sim q)) \Rightarrow r$

Open solution

Q84. Numerical · **Mathematics**

The sum and product of the mean and variance of a binomial distribution are 82.5 and 1350 respectively. Then the number of trials in the binomial distribution is _____.

Open solution

Q85. Numerical · **Mathematics**

If $\sum_{k=1}^{10} k^2 \left({}^{10}C_k\right)^2 = 22000L$, then L is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a, b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point (2, −1), then $|2a + 9b|$ is equal to _____.

Open solution

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Q87. MCQ (Single Correct Answer) · **Physics**

Open solution

Read the following statements :

(A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass number.

(D) Density of the nucleus is directly proportional to the cube root of the mass number.

(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options.

(A) (A) and (D) only.

(B) (A) and (E) only.

(C) (B) and (E) only.

(D) (A) and (C) only.

Q88. MCQ (Single Correct Answer) · **Physics**

Open solution

The vertical component of the earth's magnetic field is $6 \times 10^{-5}\ \mathrm{T}$ at any place where the angle of dip is 37°. The earth's resultant magnetic field at that place will be $\left(\text{Given } \tan 37^{\circ} = \frac{3}{4}\right)$

(A) $8 \times 10^{-5}\ \mathrm{T}$ **(B)** $6 \times 10^{-5}\ \mathrm{T}$ **(C)** $5 \times 10^{-4}\ \mathrm{T}$ **(D)** $1 \times 10^{-4}\ \mathrm{T}$

Q89. Numerical · **Physics**

Open solution

A modulating signal $2\sin\left(6.28 \times 10^{6}\right)t$ is added to the carrier signal $4\sin\left(12.56 \times 10^{9}\right)t$ for amplitude modulation. The combined signal is passed through a non-linear square law device. The output is then passed through a band pass filter. The bandwidth of the output signal of band pass filter will be _____ MHz.

Q90. Numerical · **Physics**

Open solution

Two radioactive materials A and B have decay constants 25λ and 16λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of B to that of A will be "e" after a time $\frac{1}{a\lambda}$. The value of a is _____.

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Answer Key

Chemistry

1. **C**	2. **B**	3. **D**	4. **B**	5. **C**
6. **B**	7. **A**	8. **A**	9. **C**	10. **C**
11. **C**	12. **A**	13. **D**	14. **C**	15. **4**
16. **80**	17. **34**	18. **1**	19. **9**	20. **6**
21. **3**				

Mathematics

22. **D**	23. **C**	24. **C**	25. **D**	26. **A**
27. **A**	28. **A**	29. **B**	30. **B**	31. **B**
32. **B**	33. **B**	34. **C**	35. **C**	36. **16**
37. **10**	38. **6**	39. **79**	40. **72**	41. **14**
42. **27**				

Physics

43. **B**	44. **B**	45. **C**	46. **A**	47. **B**
48. **A**	49. **D**	50. **D**	51. **C**	52. **B**
53. **A**	54. **D**	55. **B**	56. **C**	57. **C**
58. **B**	59. **C**	60. **C**	61. **4**	62. **84**
63. **50**	64. **25**	65. **15**	66. **5**	67. **480**
68. **10**				

Out of syllabus

69. **A**	70. **C**	71. **B**	72. **B**	73. **C**
74. **D**	75. **80**	76. **1**	77. **4**	78. **B**
79. **B**	80. **A**	81. **B**	82. **D**	83. **B**
84. **96**	85. **221**	86. **195**	87. **B**	88. **D**
89. **2**	90. **9**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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